NKOSINATHI TSHABALALA

SOFTWARE ENGINEER | C# / .NET 10 | AWS SERVERLESS | FULL STACK

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PROFILE

Software engineer focused on C#/.NET backend development and AWS serverless applications. Builds full-stack products, reusable platforms and event-driven systems using Clean Architecture, automated testing and infrastructure as code.

SELECTED PROJECTS

Iaac Commerce — Saelune Active [Live storefront]

Reusable ecommerce platform · React, TypeScript, .NET 10, AWS

- Built React/TypeScript storefront and admin applications with a .NET 10 serverless API and AWS CDK infrastructure.
- Designed DynamoDB tables and indexes for catalog, inventory, customers and orders; implemented variant browsing, profiles, carts, favourites and product administration.
- Developed checkout with atomic stock reservations and idempotency; integrated Peach Payments hosted checkout and Bob Go delivery.
- Used S3/CloudFront, Cognito, SES and SQS for hosting, identity, transactional email and background processing.

Primary Onboarding [Live platform]

Multi-tenant business operations SaaS · React, Next.js, .NET 10, AWS

- Built a South African freelancer/SME platform for customers, quotes, service agreements, invoicing, recurring email schedules, payment/receipt tracking, expenses and reports.
- Developed a static React/Next.js frontend and ASP.NET Core API on Lambda/API Gateway; modelled tenant- and branch-scoped access across eight DynamoDB tables.
- Integrated Cognito, SES, SQS, EventBridge Scheduler and private S3 storage; provisioned infrastructure with TypeScript CDK.
- Added GitHub Actions CI, xUnit, DynamoDB Local integration tests and CDK infrastructure assertions.

Serverless Blog & Custom CMS [Live site]

Reusable client publishing platform · C#, .NET 10, React/Next.js, AWS

- Built a reusable CMS for article drafting, review, editorial approval, scheduling and publishing, with administrator/editor roles.
- Implemented Markdown/rich-text authoring, media uploads, thumbnails, layouts, categories, search, SEO metadata, archives and sitemaps.
- Added newsletter confirmation and campaigns, subscriber management, contact forms, admin inbox and branding controls.
- Deployed client-specific AWS resources using CDK, with S3/CloudFront hosting and caching, Cognito identity, SES and SQS.

FLAGSHIP BACKEND PLATFORM

PGLN.Auth — reusable .NET 10 authentication platform · NuGet v0.1.0

- Built ASP.NET Core authentication workflows using Clean Architecture, CQRS, vertical slices, EF Core and PostgreSQL.
- Implemented registration, email verification, JWT access tokens, refresh-token rotation with replay detection, password recovery, sessions, step-up authentication and trusted devices.
- Implemented transactional outbox, SQS/Lambda/SES email processing and durable PostgreSQL inbox for idempotent handling.
- 670 automated tests passed locally across the project suite; GitHub Actions CI, PostgreSQL and Docker-based development.

TECHNICAL SKILLS

Languages & frameworks
C# · .NET 10 · ASP.NET Core · Minimal APIs · React · Next.js · TypeScript · REST

AWS & infrastructure
Lambda · API Gateway · DynamoDB · S3 · CloudFront · Cognito · SQS · SES · EventBridge · CDK · CloudFormation

Data & integration
PostgreSQL · SQL Server · EF Core · DynamoDB · JSON · XML · Peach Payments · Bob Go

Engineering
Clean Architecture · CQRS · Vertical Slice · DI · Event-driven systems · Outbox/Inbox · Idempotency · IaC

Testing & delivery
xUnit · Integration testing · DynamoDB Local · GitHub Actions · Docker · Git · Postman

OPEN-SOURCE LIBRARIES

PGLN.Media
Modular .NET media library with file-signature validation, MIME/extension checks, upload limits, storage abstractions and Azure storage integration.

SAValidation
South African phone-number validation and normalization, operator detection, landline area identification and structured validation results.

PROJECT-BASED INTEGRATION EXPERIENCE

Hands-on integrations delivered through personal and client projects: REST/JSON APIs, XML, Amazon SQS, AWS Lambda, SES, Cognito, Peach Payments and Bob Go. Experience is project-based.

EDUCATION

BSc Information Technology
Software Engineering · Eduvos, Pretoria
2022–2026
76% average · 18 distinctions

COMMUNITY

Co-founder — EduDev coding club
Co-founded a student coding club; organized weekly programming challenges and peer code reviews.

LINKS

GitHub profile
Portfolio
PGLN.Auth repository